

Technical SEO Development Requirements

1. Semantic HTML & Heading Structure

- Use semantic HTML5 tags (<header>, <nav>, <main>, <section>, <article>, <footer>).
- Ensure only one <h1> per page.
- Maintain a logical heading hierarchy (H1 → H2 → H3 → H4).
- Do not use headings for styling purposes.
- All important page content must be present in the initial HTML response and remain crawlable by search engines.

2. URL Routing & Canonicalization

- Generate clean, human-readable URLs.
- URLs should be lowercase and use hyphens (-) instead of underscores.
- Support custom slug generation from the CMS.
- Automatically generate canonical tags for every indexable page.
- Implement 301 redirects when URLs are modified.
- Avoid duplicate URL variations (trailing slash, uppercase, query parameters).

3. Metadata Management

- Metadata must be dynamically generated and editable through the CMS.
- Support:
 - Meta title
 - Meta description
 - Canonical URL
 - Meta robots
 - Open Graph tags
 - Twitter Card tags
- Generate default metadata when values are not explicitly provided.

4. Structured Data (JSON-LD)

- Implement JSON-LD schema injection.
- Support dynamic schema for:
 - BreadcrumbList
 - FAQPage
 - Organization
 - Service
 - More Schema are there to implement
- Schema values should be populated dynamically from CMS data.

5. Performance & Core Web Vitals

- Optimize for Core Web Vitals:
 - $LCP \leq 2.5$ seconds
 - $INP \leq 200$ ms
 - $CLS \leq 0.1$
- Implement:
 - Lazy loading for images and iframes
 - Code splitting
 - Tree shaking
 - CSS/JS minification
 - Brotli/Gzip compression
 - Browser caching
 - Critical CSS
 - Font preloading
 - Image optimization (WebP/AVIF)

6. Crawlability & Indexing

- Auto-generate XML sitemap.
- Provide editable robots.txt.
- Apply canonical tags to all indexable pages.
- Support index, noindex, follow, and nofollow directives.
- Return correct HTTP status codes (200, 301, 302, 404, 410).
- Generate custom 404 page.
- Ensure all internal links use crawlable `<a href="">` elements.

7. Media Optimization

- Automatically generate responsive images (srcset and sizes).
- Convert uploaded images to WebP/AVIF where supported.
- Compress images during upload.
- Support editable alt attribute for every image.
- Lazy-load non-critical images.
- Preserve original image dimensions to prevent layout shifts.

8. Responsive Design & Accessibility

- Follow a mobile-first responsive approach.
- Ensure compatibility across major browsers.
- Meet WCAG 2.1 AA accessibility guidelines where feasible.
- Use ARIA attributes only when semantically required.
- Ensure full keyboard navigation.
- Maintain adequate color contrast.
- Prevent cumulative layout shifts during rendering.

9. CMS SEO Configuration

Expose the following fields in the CMS:

- SEO title
- Meta description
- URL slug
- Canonical URL
- Meta robots
- Open Graph data
- Twitter Card data
- JSON-LD schema fields
- Image alt text
- Redirect management (301/302)

All SEO-related settings should be editable without requiring code changes.

10. Analytics & Third-Party Integration

- Integrate Google Tag Manager, Support Google Analytics 4 event tracking. Provide Google Search Console verification support. Support Microsoft Clarity integration. Allow Meta Pixel and other tracking scripts through GTM or CMS configuration. Ensure third-party scripts load asynchronously or are deferred to minimize performance impact.